



Artificial intelligence and adaptive response to market changes: A strategy to enhance firm performance and innovation

Yulia Sullivan^a, Samuel Fosso Wamba^{b,*}

^a Hankamer School of Business, Baylor University, One Bear Place #98005, Foster Campus, Waco, TX, USA

^b TBS Business School, France 1 Place Alphonse Jourdain, 31068 Toulouse, France

ARTICLE INFO

Keywords:

Artificial intelligence
Adaptive response to market changes
Innovation
Firm performance
AI-powered capabilities

ABSTRACT

This research examines how AI-powered capabilities can bring value to organizations by enhancing their Adaptive Response to Market Changes (ARMC). Utilizing insights from organizational agility and the dynamic capability framework, we define ARMC as an organization's ability to promptly identify and adjust to market changes, with customer responsiveness and operational adjustment as foundational competencies. We outline three AI-powered capabilities (AI-enabled automation, AI-enabled analytics, and AI-enabled relational capabilities) as ARMC's predictors. We posit that the strengths of these relationships depend on environmental hostility and dynamism. Additionally, we propose positive associations between ARMC and three organizational outcomes: firm performance, process innovation, and product innovation. Our research employs a two-stage design, surveying IT and business executives from firms that have adopted AI. The results demonstrate significant interaction effects of environmental hostility and dynamism on the relationships between AI-powered capabilities and ARMC. Furthermore, we find that ARMC positively influences firm performance and innovation.

1. Introduction

The rise of digitization, characterized by the profound use of data analytics and artificial intelligence (AI), has compelled organizations to adopt a new perspective on their IT assets and focus on a more intricate level (Benner & Tushman, 2015). Typically defined as “the ability of machines to perform human-like cognitive tasks,” AI is currently viewed as the most important and disruptive new technology for organizations (Benbya et al., 2020, p. ix). However, the management of AI differs significantly from previous approaches to IT management (Berente et al., 2021). AI cannot be viewed merely as a technology or a collection of technologies; rather, it represents an ever-evolving system encompassing emerging computational capabilities (Berente et al., 2021). Thus, today's leaders must navigate both the opportunities and the obstacles that come with the pervasive adoption of AI.

Further, the process of digitalization enabled by AI has triggered a paradigm shift in the fundamental theories that explain how IT innovation generates value (Nambisan et al., 2017). While the challenges of utilizing IT to create business value are fairly well understood (Barua et al., 2004; Côte-Real et al., 2017; Grover et al., 2018; Kohli & Grover, 2008; Melville et al., 2004; Wang et al., 2012), the recent rise of AI

introduces a new set of challenges to this longstanding issue. Consequently, more contemporary scrutiny and analysis is needed (Nambisan et al., 2017). The emergence of new AI capabilities and their associated challenges raises the question of how firms can effectively harness these technological advancements to gain a competitive edge in their industries (Iansati & Lakhari, 2020).

Although prior studies have shown the promising role of AI in supporting decision-making processes in an organization (e.g., Duan et al., 2019; Gupta et al., 2022; Jussupow et al., 2021; Shrestha et al., 2019), theoretical consensus is limited on how AI influences firms' responses and adaptability to market changes. However, AI can empower organizations to respond to dynamic market changes effectively (Narayandas & Sengupta, 2023). For example, it can automate and optimize various operation processes, enhancing the efficiency and responsiveness of an organization (Benbya et al., 2020; Huang & Rust, 2021). AI-enabled predictive analytics allow organizations to identify emerging market trends and connect with customers, allowing them to proactively adapt their strategies and offerings (Huang & Rust, 2021). Moreover, by leveraging AI, businesses can gain valuable insights from vast volumes of data and make informed decisions in real time (Davenport et al., 2020). In sum, AI provides organizations with advanced computing

* Corresponding author.

E-mail addresses: yulia_sullivan@baylor.edu (Y. Sullivan), s.fosso-wamba@tbs-education.fr (S. Fosso Wamba).

<https://doi.org/10.1016/j.jbusres.2024.114500>

Received 10 March 2022; Received in revised form 29 December 2023; Accepted 4 January 2024

Available online 13 January 2024

0148-2963/© 2024 Elsevier Inc. All rights reserved.